

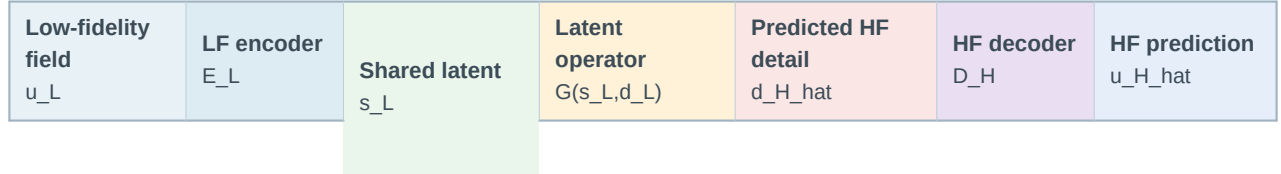
Shared-Detail Latent Neural Operator

A proposed architecture for multi-fidelity PDE field prediction

Executive summary

The central proposal is to decompose low- and high-fidelity PDE fields into two kinds of latent information: **shared physics** that should agree across fidelities, and **fidelity-specific detail** that should remain distinct. A latent neural operator then predicts only the missing high-fidelity detail rather than reconstructing the entire high-fidelity field from scratch.

This combines four established ideas - latent neural operators, multi-fidelity residual learning, shared/private representation learning, and physics-aware frequency decomposition - in a specific formulation aimed at high-frequency PDE solutions.



Core hypothesis: LF and HF simulations should agree on large-scale physical structure, but HF contains additional information that must not be destroyed by forcing the complete latent representations to match.

Why not align the full LF and HF latent?

If training enforces $E_L(u_L) = E_H(u_H)$ everywhere, the latent operator becomes approximately the identity. Worse, the HF encoder may discard exactly the fine-scale information the model is supposed to predict. The proposed split avoids this by aligning only the shared component.

$$E_L(u_L) = (s_L, d_L) \text{ and } E_H(u_H) = (s_H, d_H)$$

Align s_L with s_H , but do not align d_L with d_H .

1. Model architecture

Latent fields rather than global vectors

A latent vector compresses an entire field into one global code. A latent field retains spatial organization, such as a $64 \times 64 \times C$ feature map. For PDEs, preserving location matters because a vortex, shock, or boundary layer must remain tied to where it occurs. The latent operator therefore acts on a compressed spatial field, not a single vector.

Component	Role	Suggested implementation
LF encoder E_L	Extract shared structure and LF-specific cues	CNN, FNO encoder, graph encoder, or patch/token encoder
HF encoder E_H	Provide training targets s_H and d_H	Parallel architecture to E_L ; used during training
Latent operator G	Predict missing HF detail	Latent FNO first; Mamba or Transformer as later variants
HF decoder D_H	Combine shared physics and predicted HF detail	Upsampling decoder with skip connections if needed

Prediction pathway

$$d_H_{\text{hat}} = G(s_L, d_L, \text{theta})$$

$$u_H_{\text{hat}} = D_H(s_L, d_H_{\text{hat}})$$

The conditioning variables theta may include geometry, Reynolds number, boundary conditions, forcing parameters, or other PDE descriptors. Although d_L is fidelity-specific, it can still contain useful clues about numerical artifacts or unresolved structures, so the operator should generally receive both s_L and d_L .

Recommended physical interpretation

Shared latent s : large-scale flow topology, dominant vortices, wake location, shock position, low-frequency pressure structure.

Detail latent d : fine vortices, boundary-layer thickness, high-wavenumber content, solver-specific corrections, and HF-only structure.

2. Training objectives

A channel split alone is insufficient because the network can hide information in either branch. The split must be anchored by several complementary losses.

HF prediction

$$L_{\text{HF}} = ||u_{\text{H_hat}} - u_{\text{H}}||^2$$

Ensures the full model solves the downstream task.

Autoencoder reconstruction

$$L_{\text{rec}} = ||D_L(s_L, d_L) - u_L||^2 + ||D_H(s_H, d_H) - u_H||^2$$

Prevents the latent from discarding necessary information.

Shared alignment

$$L_{\text{shared}} = ||s_L - s_H||^2$$

Aligns only the information genuinely common across fidelities.

Detail prediction

$$L_{\text{detail}} = ||G(s_L, d_L) - \text{stopgrad}(d_H)||^2$$

Makes the operator target HF-specific information.

Separation

$$L_{\text{cov}} = ||\text{Cov}(s, d)||_F^2$$

Discourages shared and detail channels from duplicating one another.

Frequency-guided identifiability

The most practical way to give the split a concrete meaning is to make the shared-only decoder reconstruct a low-pass target, while the detail branch reconstructs the remaining correction.

$$D_H(s_H, \theta) \approx \text{LP}(u_H)$$

$$D_{\text{detail}}(d_H) \approx \text{HP}(u_H - U(u_L))$$

Here LP and HP denote low- and high-pass operations, and U upsamples the LF field onto the HF grid. This should be treated as a soft inductive bias rather than a rigid claim that all shared information is low-frequency and all detail is high-frequency.

Optional stronger constraints

- **Cross-reconstruction:** $D_H(s_L, d_H)$ should reconstruct the paired HF field.
- **Fidelity adversary:** a classifier should fail to infer fidelity from s , but may infer it from d .
- **Spatial-capacity bias:** use a coarser shared latent and a finer-resolution detail latent.
- **Multi-step refinement:** predict detail progressively across frequency bands or resolutions.

3. Relation to prior work and expected novelty

The proposal is best presented as a new synthesis rather than as an entirely unprecedented family. Prior work has separately explored each neighboring ingredient.

Prior direction	What it contributes	What remains different here
Latent neural operators	Encode, operate in latent space, decode	Usually treat the latent primarily as compression, not an explicit shared/detail decomposition
Shared multi-fidelity latent spaces	Align LF and HF representations	Full alignment can erase HF-only information and make the operator trivial
Residual MF operators	Predict LF-to-HF correction	Usually predict a physical-space residual rather than an HF-private latent
Shared/private representation learning	Separate invariant and domain-specific factors	Mostly developed outside PDE operator learning

Safest novelty claim

Prior work has separately studied shared latent spaces across fidelities, residual LF-to-HF operator learning, latent-space PDE models, and shared/private representation decomposition. The proposed contribution is to combine these by explicitly decomposing multi-fidelity PDE fields into aligned shared and fidelity-specific latent fields and training a latent neural operator to predict only the high-fidelity-specific component.

Why this could outperform a regular neural operator

- The operator does not need to relearn large-scale physics already present in LF.
- Computation occurs on a smaller latent field rather than the full-resolution grid.
- The detail branch can allocate capacity specifically to high-frequency corrections.
- The decomposition may improve data efficiency when HF samples are scarce.

Main failure modes

- The encoder discards fine scales before the operator sees them.
- The shared branch absorbs all information and the detail branch collapses.
- The detail branch memorizes sample identity or solver artifacts.
- An overly strong alignment loss removes real LF/HF differences.
- The added model complexity outweighs any benefit over an end-to-end FNO.

4. Recommended first experiment

Baseline architecture

Start with a latent FNO rather than Mamba. This isolates whether the shared/detail representation helps before changing the operator family.

Stage	Model	Purpose
A	Regular LF-to-HF FNO	Main end-to-end baseline
B	Plain autoencoder + latent FNO	Measures benefit of compression alone
C	Latent residual operator	Predicts one undivided HF latent correction
D	Frequency-guided shared/detail latent FNO	Tests the proposed decomposition
E	D + covariance/cross-reconstruction losses	Tests learned disentanglement beyond frequency bias

Evaluation

- **Field error:** relative L2, normalized RMSE, and worst-case error.
- **Spectral error:** error by wavenumber band, especially the high-frequency tail.
- **Structure:** vortex, shock, or boundary-layer location and intensity.
- **Efficiency:** parameters, memory, inference time, and training cost.
- **Data efficiency:** performance as the number of paired HF samples decreases.
- **Robustness:** unseen PDE parameters, resolutions, geometries, and solver settings.

Latent ablations that verify the split

Decoder input	Expected behavior
s_H only	Correct broad field but missing sharp detail
d_H only	Fine texture or residual statistics without correct global structure
s_H + d_H	Accurate full HF reconstruction
s_L + d_H	Nearly reconstructs the paired HF field
s_H + shuffled d_H	Global structure follows s_H; fine statistics change with d_H

Bottom line

The strongest initial version is a frequency-guided shared/detail latent FNO: align shared LF/HF latents, make shared-only decoding reconstruct the coarse HF component, and train the latent operator to predict the missing HF-detail latent. Only after this version demonstrates value should the project add Mamba, adversarial invariance, or autoregressive multi-stage refinement.